Verona through Rome

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1 Introduction

We present $Roma^M$, a more complete, fully formalized version of $Rome^M$ [Nguyen 2026].

2 Model

The configuration remains unchanged from the report. We make one change to the semantics, specifically within `makeRegion`:

$$\frac{\begin{array}{c}\text{MAKE-REGION}\\ F = (rid, bvar, omap, vmap, fid) \\ \langle (S :: F), H\rangle.freshObjectId = oid' \qquad \langle (\mathbf{S} :: \mathbf{F}), \mathbf{H}\rangle.\mathbf{freshRegionId} = \mathbf{rid}' \\ R' = (oid', [oid' \mapsto \emptyset], Closed) \qquad F' = (rid, bvar, omap, vmap[\mathtt{x} \mapsto \mathsf{RId}(rid')], fid)\end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeRegion x}} \langle S :: F', H[rid \mapsto R']\rangle}$$

In the report, the `regionId` would coincide with the `objectId` of the bridge object, but here we've chosen to completely decouple them. The original reasoning behind this was to future-proof for deallocation, but in writing this we found out that the argument didn't hold, and it would've been fine to leave `makeRegion` as before. Luckily, this doesn't change much, it only means that we need to prove regions are unique separately from objects, as opposed to using a combined argument before.

3 Invariants

We add two new invariants, presented here informally and formally.

Definition 3.1 (HS1). Every reference to an object must be to an object that exists in the configuration.

$$\forall oid : [\mathsf{OId}(oid) \in \langle S, H\rangle.refs \implies oid \in \langle S, H\rangle.objectIds]$$

Definition 3.2 (HS2). Every reference to a region must be to a region that exists in the configuration.

$$\forall rid : [\mathsf{RId}(rid) \in \langle S, H\rangle.refs \implies rid \in dom(H)]$$

These came up in the process of proving *S2* and *S3* for `makeObjStack`, `makeObjRegion`, and `makeRegion`. This stems from the fact that our `freshObjectId` and `freshRegionId` functions only take into account the `Ids` used as keys when creating a new `Id`. Thus, without this invariant, it is technically possible[1] for a configuration to contain a dangling reference to an `objectId` or `regionId` that does not exist within the configuration, but is then inserted via one of the three `make` operations.

[1]Although it isn't possible within the context of an operational semantics starting with the starting configuration, which is why we were able to prove it without altering the operational semantics.

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4 Reachability

Definition 4.1 (Stack-reachable). An object or region is stack-reachable if it is frame-reachable from any frame in the stack.

$$\frac{\langle S, H\rangle \vdash FReachable(\_, a)}{\langle S, H\rangle \vdash SReachable(a)}$$

Definition 4.2 (Frame-reachable). An object or region is frame-reachable if

$$\textsc{Var}\ \frac{F = (\_, \_, \_, vmap, fid) \qquad vmap(\_) = a}{\langle S :: F, H\rangle \vdash FReachable(fid, a)} \qquad \textsc{Bridge}\ \frac{F = (rid, \_, \_, \_, fid) \qquad H(rid) = (boid, \_, \_)}{\langle S :: F, H\rangle \vdash FReachable(fid, \mathsf{OId}(boid))}$$

$$\textsc{Step}\ \frac{\langle S, H\rangle \vdash FReachable(fid, a) \qquad \langle S, H\rangle \vdash a \longrightarrow b}{\langle S, H\rangle \vdash FReachable(fid, b)}$$

Definition 4.3 (Region-reachable). An object or region is region-reachable if

$$\textsc{Bridge}\ \frac{H(rid) = (boid, \_, \_)}{\langle S, H\rangle \vdash RReachable(rid, \mathsf{OId}(boid))} \qquad \textsc{Step}\ \frac{\langle S, H\rangle \vdash RReachable(rid, a) \qquad \langle S, H\rangle \vdash a \longrightarrow b}{\langle S, H\rangle \vdash RReachable(rid, b)}$$

Definition 4.4 (Reachable-step).

$$\textsc{Obj-Step}\ \frac{\mathsf{OId}(oid).objAt(\langle S, H\rangle) = obj \qquad b \in obj.refs}{\langle S, H\rangle \vdash \mathsf{OId}(oid) \longrightarrow b} \qquad \textsc{Region-Step}\ \frac{H(rid) = (boid, \_, Closed)}{\langle S, H\rangle \vdash \mathsf{RId}(rid) \longrightarrow \mathsf{OId}(boid)}$$

THEOREM 4.5.

$$\langle S, H\rangle \vdash RReachable(rid, a) \iff \exists boid : [H(rid) = (boid, \_, \_) \land \mathsf{OId}(boid) \xrightarrow{*} a]$$

THEOREM 4.6.

$$\langle S, H\rangle \vdash FReachable(fid, a) \iff \exists start : [\langle S, H\rangle \vdash FRoot(fid, start) \land start \xrightarrow{*} a]$$

Definition 4.7 (Frame-root).

$$\textsc{Var}\ \frac{F = (\_, \_, \_, vmap, fid) \qquad vmap(\_) = a}{\langle S :: F, H\rangle \vdash FRoot(fid, a)} \qquad \textsc{Bridge}\ \frac{F = (rid, \_, \_, \_, fid) \qquad H(rid) = (boid, \_, \_)}{\langle S :: F, H\rangle \vdash FRoot(fid, \mathsf{OId}(boid))}$$

5 Related Work

6 Conclusion

A Configuration

$$
\begin{aligned}
RegionId, ObjectId, Index &\subseteq Int \\
VarName, FieldName &\subseteq String \\
\\
RuntimeConfiguration &\triangleq Heap \times Stack \\
\\
Heap &\triangleq RegionId \rightharpoonup Region \\
Region &\triangleq ObjectId \times ObjMap \times Status \\
Status &\triangleq Open \mid Closed \\
\\
Stack &\triangleq List\ Frame \\
Frame &\triangleq RegionId \times VarName \times ObjMap \times VarMap \times Index \\
\\
ObjMap &\triangleq ObjectId \rightharpoonup FieldMap \\
\\
Object &\triangleq FieldMap \\
FieldMap &\triangleq FieldName \rightharpoonup Reference \\
VarMap &\triangleq VarName \rightharpoonup Reference \\
\\
Reference &\triangleq \mathsf{RId}(RegionId) \mid \mathsf{OId}(ObjectId)
\end{aligned}
$$

B Operational Semantics

VAR-ASGN-STACK

$$
\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \epsilon \qquad resolveFA(\mathsf{y}\,.\,\mathsf{f}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x\ :=\ y.f}} \langle S :: (rid, bvar, omap, vmap[\mathsf{x} \mapsto \mathsf{OID}(oid)], fid), H\rangle}
$$

VAR-ASGN-BRIDGE

$$
\frac{\begin{array}{c} F = (rid, \mathsf{x}, \_, \_, \_) \qquad resolveFA(\mathsf{y}\,.\,\mathsf{f}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \\ \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(rid) = (boid, omap, status) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\mathtt{x\ :=\ y.f}} \langle (S :: F), H[rid \mapsto (oid, omap, status)]\rangle}
$$

$$\textsc{field-asgn-stack}\quad \frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \qquad \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Stk}(fid) \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid') \qquad omap(oid) = obj \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: (rid, bvar, omap[oid \mapsto obj[\mathsf{f} \mapsto OId(oid')]], vmap, fid)), H\rangle}$$

$$\textsc{field-asgn-region}\quad \frac{\begin{array}{c} F = (rid, \_, \_, \_, \_) \qquad resolveV(\mathsf{x}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid) \\ \mathsf{OId}(oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(oid') \\ \mathsf{OId}(oid').loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(rid) = (boid, omap, Open) \qquad omap(oid) = obj \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: F), H[rid \mapsto (boid, omap[oid \mapsto obj[\mathsf{f} \mapsto \mathsf{OId}(oid')]], Open)]\rangle}$$

$$\textsc{swap-stack}\quad \frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{x}ref \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \\ resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{yf}ref \qquad \mathsf{OId}(\mathsf{y}oid).loc(\langle (S :: F), H\rangle) = \mathsf{Stk}(fid) \\ omap(\mathsf{y}oid) = obj \qquad omap' = omap[\mathsf{y}oid \mapsto obj[\mathsf{f} \mapsto \mathsf{x}ref]] \\ vmap' = vmap[\mathsf{x} \mapsto \mathsf{yf}ref] \qquad F' = (rid, bvar, omap', vmap') \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: F'), H\rangle}$$

$$\textsc{swap-region-object}\quad \frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \\ \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{OId}(\mathsf{x}oid) \qquad resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \\ resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{yf}ref \qquad \{\mathsf{OId}(\mathsf{x}oid), \mathsf{OId}(\mathsf{y}oid)\}.loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \qquad obj' = obj[\mathsf{f} \mapsto \mathsf{OId}(\mathsf{x}oid)] \\ R' = (boid, romap[\mathsf{y}oid \mapsto obj'], Open) \qquad F' = (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{yf}ref]) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: F'), H[rid \mapsto R']\rangle}$$

$$\textsc{swap-region-region}\quad \frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad \mathsf{x} \neq bvar \qquad vmap(\mathsf{x}) = \mathsf{RId}(\mathsf{x}rid) \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{yf}ref \\ \mathsf{OId}(\mathsf{y}oid).loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \qquad H(xrid) = (\_, \_, Closed) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \qquad obj' = obj[\mathsf{f} \mapsto \mathsf{RId}(\mathsf{x}rid)] \\ R' = (boid, romap[\mathsf{y}oid \mapsto obj'], Open) \qquad F' = (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{yf}ref]) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: F'), H[rid \mapsto R']\rangle}$$

$$\textsc{swap-region-bridge}\quad \frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad \mathsf{x} = bvar \\ resolveV(\mathsf{y}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{y}oid) \qquad resolveFA(\mathsf{y.f}, \langle (S :: F), H\rangle) = \mathsf{OId}(\mathsf{yf}oid) \\ \{\mathsf{OId}(\mathsf{y}oid), \mathsf{OId}(\mathsf{yf}oid)\}.loc(\langle (S :: F), H\rangle) = \mathsf{Rgn}(rid) \\ H(rid) = (boid, romap, Open) \qquad romap(\mathsf{y}oid) = obj \\ obj' = obj[\mathsf{f} \mapsto \mathsf{OId}(boid)] \qquad R' = (\mathsf{yf}oid, romap[\mathsf{y}oid \mapsto obj'], Open) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{x.f := y}} \langle (S :: F), H[rid \mapsto R']\rangle}$$

ENTER

$$\frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad resolveFA(\texttt{x.f}, \langle S :: F, H\rangle) = \mathsf{RId}(rid') \\ H(rid') = (boid, romap, Closed) \qquad F' = (rid', \mathsf{a}, \emptyset, \emptyset, fid+1) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{enter x.f a}} \langle (S :: F :: F'), H[rid' \mapsto (boid, romap, Open)]\rangle}$$

EXIT

$$\frac{F = (rid, \_, \_, \_, \_) \qquad H(rid) = (boid, omap, Open)}{\langle (S :: F' :: F), H\rangle \xrightarrow{\texttt{exit}} \langle S :: F', H[rid \mapsto (boid, omap, Closed)]\rangle}$$

MAKE-OBJ-STACK

$$\frac{F = (rid, bvar, omap, vmap, fid) \qquad \langle (S :: F), H\rangle.freshObjectId = oid}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeObjStack x}} \langle S :: (rid, bvar, omap[oid \mapsto \emptyset], vmap[\mathsf{x} \mapsto \mathsf{OId}(oid)], fid), H\rangle}$$

MAKE-OBJ-REGION

$$\frac{\begin{array}{c} F = (rid, bvar, fomap, vmap, fid) \qquad \langle (S :: F), H\rangle.freshObjectId = oid \\ H(rid) = (boid, romap, Open) \qquad R = (boid, romap[oid \mapsto \emptyset], Open) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeObjRegion x}} \langle S :: (rid, bvar, fomap, vmap[\mathsf{x} \mapsto \mathsf{OId}(oid)], fid), H[rid \mapsto R]\rangle}$$

MAKE-REGION

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ \langle (S :: F), H\rangle.freshObjectId = oid' \qquad \langle (S :: F), H\rangle.freshRegionId = rid' \\ R' = (oid', [oid' \mapsto \emptyset], Closed) \qquad F' = (rid, bvar, omap, vmap[\mathsf{x} \mapsto \mathsf{RId}(rid')], fid) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{makeRegion x}} \langle S :: F', H[rid \mapsto R']\rangle}$$

MERGE

$$\frac{\begin{array}{c} F = (rid, bvar, omap, vmap, fid) \\ vmap(\mathsf{x}) = \mathsf{RId}(rid') \qquad H(rid) = (boid, romap, Open) \qquad H(rid') = (boid', romap', Closed) \\ F' = (rid, bvar, omap, vmap[\mathsf{x} \mapsto boid'], fid) \qquad R = (boid, romap \cup romap', Open) \end{array}}{\langle (S :: F), H\rangle \xrightarrow{\texttt{merge x}} \langle S :: F', H[rid' \mapsto \epsilon][rid \mapsto R]\rangle}$$

Definition B.1 (Name). A configuration σ satisfies P if

$$\forall x. \quad \big|\{\, y \mid y \in X(\sigma),\ y = x \,\}\big| \leq 1.$$

Definition B.2 (Name).

$$\forall x \in X(\sigma).\ \exists y.\ P(x, y) \wedge Q(y).$$

Definition B.3 (Step relation).

$$\frac{P(\sigma, a) \qquad Q(a, f, b)}{\sigma \vdash a \xrightarrow{f} b}$$

Definition B.4 (Reachable).

$$Reachable(\sigma, i, b) \iff \exists a.\ Root(\sigma, i, a) \wedge \sigma \vdash a \xrightarrow{*} b.$$

Definition B.5 (Two-branch step relation).

STEP-OBJECT

$$\frac{P(\sigma, a) = c \qquad c.f = b}{\sigma \vdash a \xrightarrow{r} b}$$

STEP-REGION

$$\frac{Q(\sigma, a) = c \qquad guard(c) \qquad b = target(c)}{\sigma \vdash a \xrightarrow{r} b}$$

Definition B.6 (Inductively-closed reachability).

$$\frac{x \in X(\sigma) \qquad P(x) = a}{Reach(\sigma, i, a)}\ \textsc{Base-A} \qquad \frac{x \in X(\sigma) \qquad Q(x) = a}{Reach(\sigma, i, a)}\ \textsc{Base-B} \qquad \frac{\sigma \vdash a \xrightarrow{f} b \qquad Reach(\sigma, i, a)}{Reach(\sigma, i, b)}\ \textsc{Step}$$

Theorem B.7 (Name). *Statement.*

Theorem B.8 (Name). *Let σ be such that $P(\sigma)$. Then*

$$Q(\sigma) \implies R(\sigma).$$

References

Hoang Nguyen. 2026. *Verona through Rome - Revisiting Verona's model and region system.* Technical Report. Imperial College London. https://github.com/hxcuber/verona-through-rome/blob/main/report.pdf